Temperature Sensor Unit Using NTC Thermistor

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Sensors and Applications Laboratory

Week 2 Report

1. Introduction

The purpose of this experiment is to design and test a temperature sensing unit using an NTC (Negative Temperature Coefficient) thermistor, whose resistance decreases with an increase in temperature. The experiment aims to convert this resistance variation into a measurable and approximately linear voltage output using an appropriate signal conditioning circuit.

It also seeks to study the behavior of the thermistor, address its non-linear characteristics through suitable techniques, and analyze the overall performance of the system through simulation and practical implementation. In addition, the experiment provides an understanding of sensor interfacing, analog signal processing, and basic data acquisition methods.

2. Experiment

Designing and analysing a temperature sensing unit using an NTC thermistor.

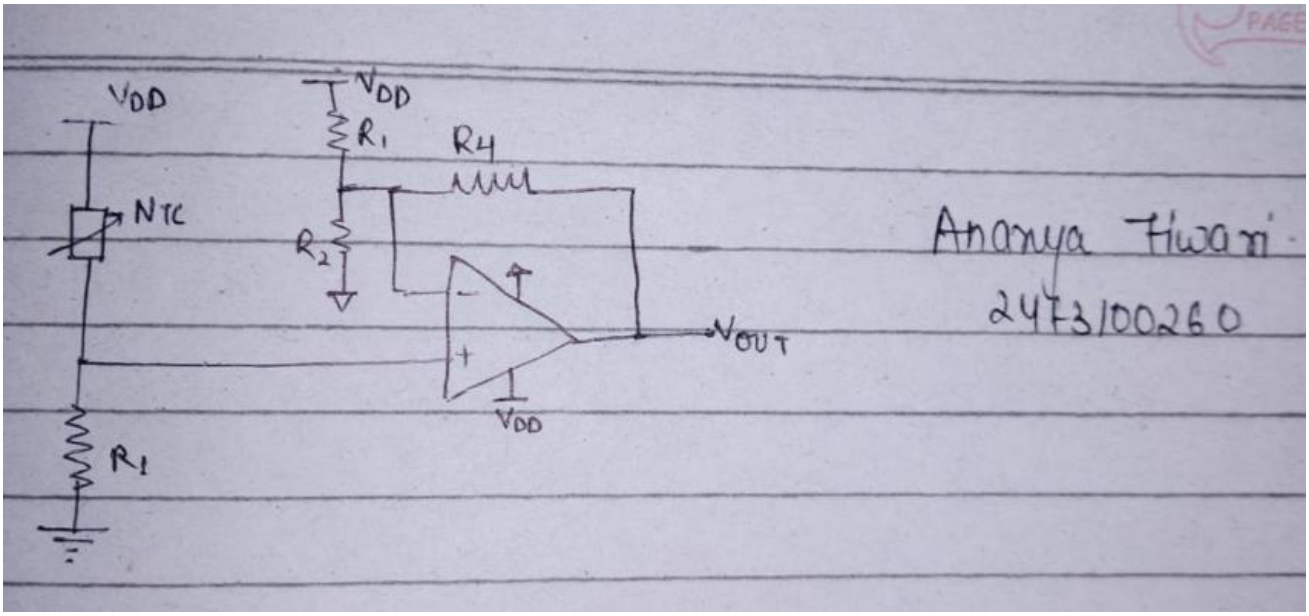


Figure 1: NTC thermistor temperature-sensing circuit with op-amp signal conditioning.

3. List of Components

- Op-Amp: MCP6004 or equivalent
- Breadboard
- Resistors
- NTC thermistor

4. Design and Procedure

4.1 Thermistor Characterization

The NTC thermistor was characterized using two measured temperature-resistance points. At 33.4°C, the thermistor resistance was 6.9 kΩ, while at 42.2°C it was 5.2 kΩ. Using a linear approximation, the slope and intercept were obtained as follows.

At temperature 33.4°C

$$\text{NTC Resistance} = 6.9\ \text{k}\Omega$$

At temperature 42.2°C

$$NTC\ Resistance = 5.2\ k\Omega$$

$$\text{Slope} = m = \frac{R_2 - R_1}{T_2 - T_1}$$

$$\text{m} = \frac{5.2 - 6.9}{42.2 - 33.4}$$

$$\text{m} = -0.1932\ \text{k}\Omega/^\circ\text{C}$$

$$R = mT + c$$

$$R = \text{-}0.1932\text{k}\Omega/^\circ\text{C} + c$$

$$6.9 = -0.1932 \times 33.4 + c$$

$$c = 13.3469$$

$$R = -0.1932T + 13.3469\ \text{k}\Omega$$

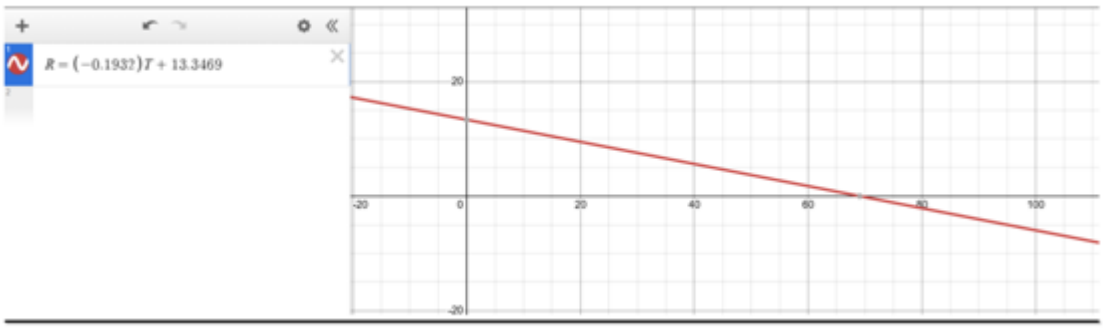


Figure 2: Linear approximation.

The resulting approximate relation is R = −0.1932T + 13.3469 kΩ.

4.2 Resistance Range

Using the linear approximation, the thermistor resistance at 60°C was obtained as 1.7662 kΩ and the resistance at 30°C as 7.5509 kΩ. These values were used as the minimum and maximum resistance limits for the circuit design.

$$R\ at\ 60°C = -0.1932 \times 60 + 13.3469$$

$$R_{min} = 1.7662\text{k}\Omega$$

$$R\ at\ 30°C = -0.1932 \times 30 + 13.3469$$

$$R_{max} = 7.5509\text{k}\Omega$$

Calculating R_1:

$$R_1 = \sqrt{(R_{min} \times R_{max})}$$

$$R_1 = \sqrt{(1.7662\text{k}\Omega \times 7.5509\text{k}\Omega)}$$

$$R_1 = 3.652\text{k}\Omega$$

$$R_1 \approx 3\text{k}\Omega$$

Calculating $V_{in(max)}$:

$$V_{in(max)} = V_{DD} \times \left(\frac{R_1}{R_{min} \times R_1}\right)$$

$$V_{in(max)} = 5\text{V} \times \left(\frac{3.652\text{k}\Omega}{1.7662\text{k}\Omega \times 3.652\text{k}\Omega}\right)$$

$$V_{in(max)} = 3.369\text{V}$$

Calculating $V_{in(min)}$:

$$V_{in(min)} = V_{DD} \times \left(\frac{R_1}{R_{max} \times R_2}\right)$$

$$V_{in(min)} = 5\text{V} \times \left(\frac{3.652\text{k}\Omega}{7.5509\text{k}\Omega \times 3.652\text{k}\Omega}\right)$$

$$V_{in(min)} = 1.63\text{V}$$

4.3 Gain Calculation

The resistor R1 was selected from the calculated resistance range. The maximum and minimum input voltages were calculated using the 5 V supply and the thermistor divider. The ideal gain required to map the input-voltage range to the desired output range was then calculated as approximately 2.53 V/V.

The calculated resistor values were R2 ≈ 2.068 kΩ, R3 ≈ 1.864 kΩ and R4 = 1.5 kΩ. The resulting actual gain was approximately 2.529 V/V.

Gain Calculation:

$$Gain_{ideal} = \frac{V_{o\,max} - V_{o\,min}}{V_{i\,max} - V_{i\,min}}$$

$$Gain_{ideal} = \frac{4.5 - 0.1}{3.369 - 1.63} \approx 2.53\ V/V$$

$$R_2 = \frac{(R_2 \parallel R_3)_{ideal} \times R_3}{R_3 - (R_2 \parallel R_3)_{ideal}}$$

$$R_2 = \frac{0.9804 \times 1.864}{1.864 - 0.9804}\ k\Omega$$

$$R_2 = 2.068\ k\Omega$$

$$Gain_{actual} = \frac{(R_2 \parallel R_3) + R_4}{(R_2 \parallel R_3)}$$

$$Gain_{actual} = \frac{0.981 + 1.5}{0.981}$$

$$Gain_{actual} = 2.529\ V/V$$

4.4 Circuit Simulation (LTspice)

The designed circuit was implemented in LTspice using the calculated resistor values. The NTC resistance was swept from 1.7662 kΩ to 7.5562 kΩ in steps of 0.1 kΩ to observe the variation in input and output voltage.

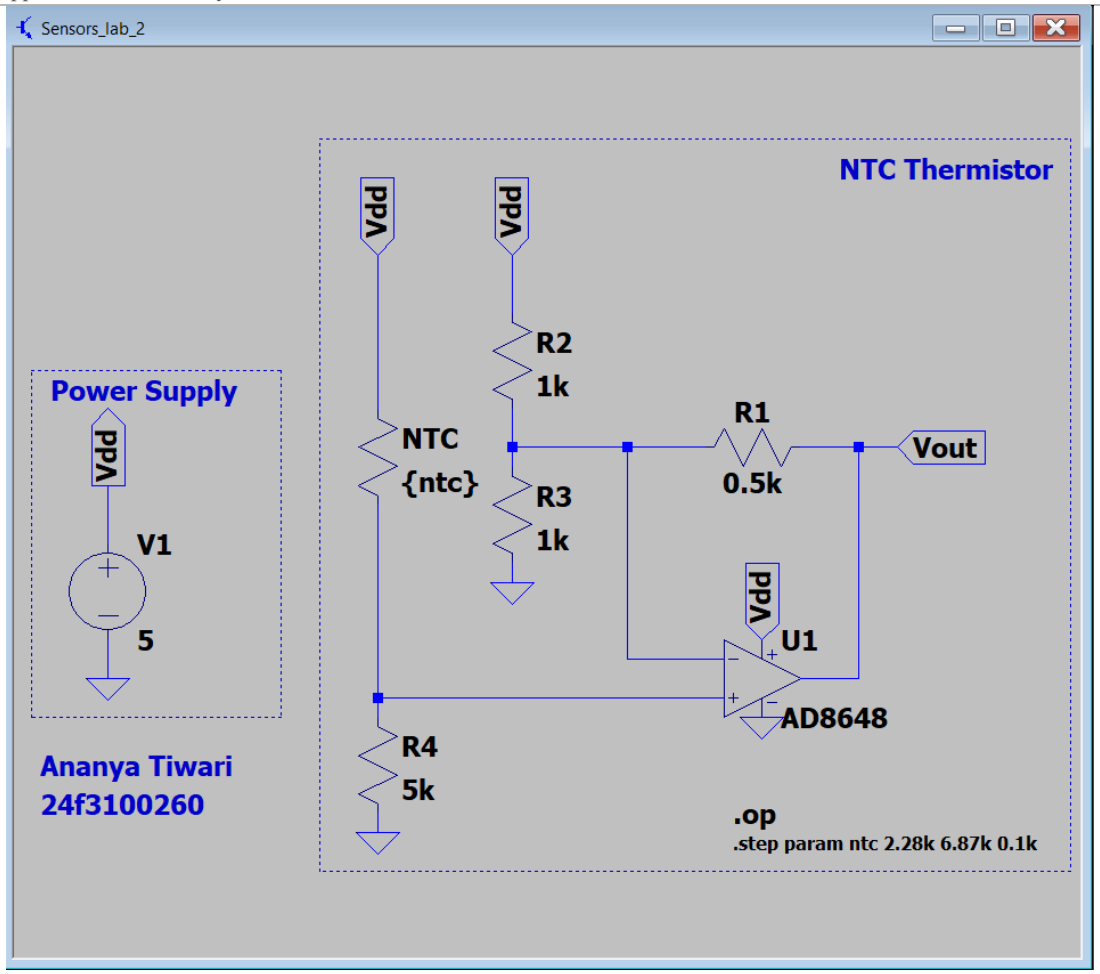


Figure 3: LTspice schematic of the NTC temperature-sensing circuit.

5. Simulation Results

The simulation was used to observe both the input voltage from the NTC divider and the conditioned output voltage. The plotted response shows the variation of Vin and Vout as the thermistor resistance changes.

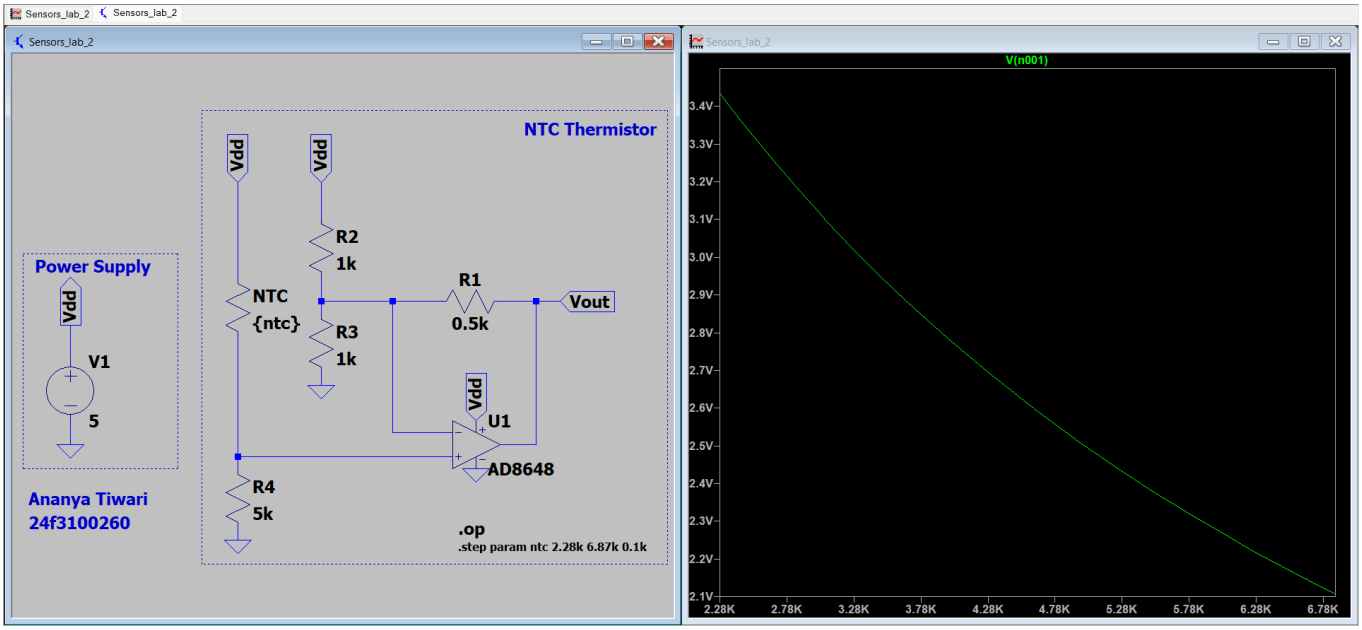


Figure 4: Simulated Vout and Vin variation with NTC resistance.

5.1 Interpretation

As the NTC resistance decreases with increasing temperature, the divider voltage changes accordingly. The op-amp signal-conditioning stage converts this changing input into a larger output-voltage variation suitable for temperature measurement.

6. Hardware Assembly

The designed circuit was assembled on a breadboard using the selected resistors, NTC thermistor and MCP6004 op-amp. The practical setup was used to measure the output voltage at different temperatures.

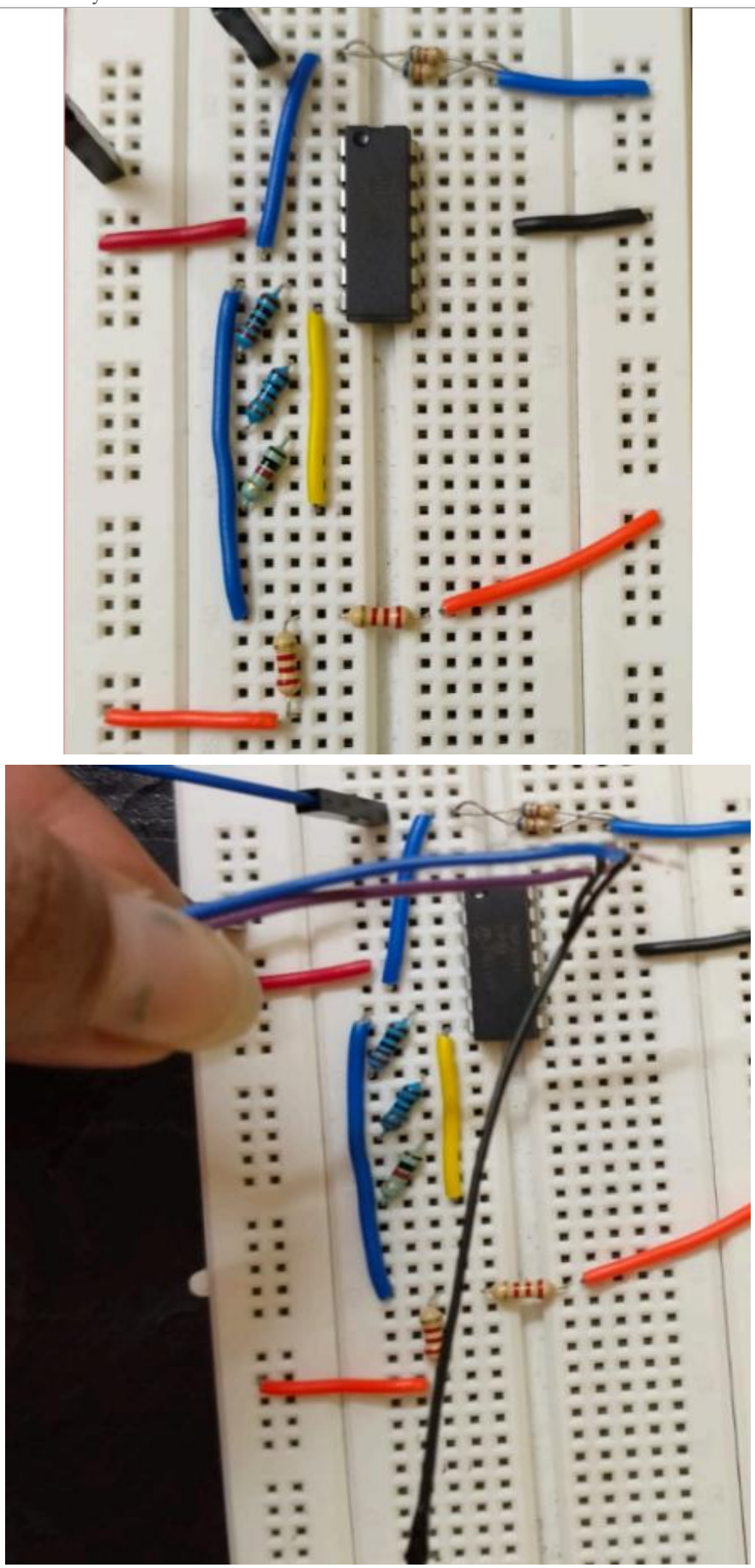


Figure 5: Breadboard implementation of the NTC temperature-sensing circuit.

7. Experimental Results

The measured temperature, thermistor resistance and output voltage were recorded at different operating points. A thermometer was used for cross-checking the temperature measurement.

Reference Temperature	Temperature	Resistance	V output	Error
31	30.6	7.4 kΩ	1.7 V	1.307189542
33	32.5	7 kΩ	1.79 V	1.538461538
35	34.8	6.3 kΩ	1.9 V	0.5747126437
37	36	5.5 kΩ	2.3 V	2.777777778
40	41	5.3 kΩ	2.5 V	-2.43902439

8. Thermometer Cross-Check



Figure 6: Thermometer cross-check.

9. Room Temperature Monitoring

Room temperature was monitored over the recorded time interval. The observed temperature varied during the measurement period.

Time	Temperature
12:00	31
01:00	32
02:00	32.5
03:00	33
04:00	34
05:00	32
06:00	30.8

10. Conclusion

This experiment showed how to design and test a temperature sensing circuit using an NTC thermistor. The circuit converted the change in resistance of the thermistor with temperature into a voltage output. The results were checked using simulation and by comparing with a thermometer. A nearly linear relationship between temperature and output was observed in the given range. However, better linearization and calibration can improve accuracy. Overall, this experiment helped in understanding sensor interfacing and basic analog circuits for temperature measurement.